



RAGHAV MADAN

Final Year Undergraduate
Department of Computer Science and Engineering
Indian Institute of Technology Kanpur

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Examination	Degree/Certificate	Institute	Year	CPI/%
Graduation	B.Tech CSE-EE	Indian Institute of Technology Kanpur	2022-2026	9.0/10
Intermediate/+2	CBSE(XII)	The Heritage School, Rohini	2022	92%
Matriculation	CBSE(X)	The Heritage School, Rohini	2020	92.2%

SCHOLASTIC ACHIEVEMENTS

- Awarded with **Academic Excellence Award** for exceptional academic performance at IIT Kanpur
- Secured **All India Rank 1298** in the Common Rank List in **JEE Advanced 2022** among 0.25 million shortlisted candidates
- Secured **All India Rank 1276** in the Common Rank List in **JEE Mains 2022** among 1 million candidates
- Received **A/A*** in 20+ courses for **exceptional performance**
- Secured **All India Rank 23** in **Srinivasa Ramanujan Mathematics Competition 2024 Prelims**, conducted by IIT Madras
- Achieved a distinguished score of 325/390 in **BITSAT 2022**, being among **top 0.1%** of the 3 Lakh applicants across the country

PROGRAMMING ACHIEVEMENTS

- Competent in **Competitive Programming** with max rating of **1639 (Expert)** on Codeforces and **1910 (4 star)** on Codechef
- Secured **Global Rank 711** in round 2 of **Meta Hacker Cup 2025**
- Achieved **Global Rank 17** in Codechef Starters 114 Div 3 and **Global Rank 417 & All India Rank 77** in Codeforces Round 954

WORK EXPERIENCE

Microsoft

- Software Engineering Intern, Windows Team May'25 – July'25
Hyderabad, India
- Engineered a **migration pipeline** to automate the **WinUI2 to WinUI3 migration process**, enabling **over 80%** of any WinUI2 codebase to be migrated to WinUI3 in a single click
 - Leveraged Microsoft's internal **Bandish Framework** to automate the migration process and make **repository-level code changes**
 - Architected a comprehensive **knowledge base of scenario-based prompts**, triggered by **regex pattern matching** on the codebase, to automatically **refactor APIs**, **update namespaces**, and **modify configuration files** for WinUI2 to WinUI3 migration
 - Successfully migrated the **WinUI2 Hardware Confirmator** to WinUI3 by leveraging the Bandish framework, reducing **over 500 lines of manual code changes** to a single click operation, and tested it by running it on a **Virtual Machine**
 - Secured a **Pre-Placement Offer** in recognition of exceeding internship expectations & delivering high-impact project results

GoalBridge AI

- Software Engineering Intern Apr'25 – May'25
Remote Intern
- Migrated an existing Google Drive Closeout Wizard from Google Apps Script to the Microsoft 365 platform by re-implementing the frontend as a client-side **SPFx Web Part** and integrating it with the backend via the **Microsoft Graph API**
 - Developed an **SPFx ListView Command Set extension** to add a 'Vectorize' option to the folder right-click menu, triggering a **backend flow** to automatically download and prepare all files within the selected folder using Microsoft Graph API for a subsequent vectorization process

TECHNICAL SKILLS

- Programming Languages** : C, C++, Python, Haskell, Ruby, Javascript, Verilog HDL, MIPS Assembly Language
- Tools and Utilities** : Git/Github, Bash Script, PowerShell Script, Vim, Vivado, HTML, $\LaTeX$

RELEVANT COURSES

Operating Systems	Probability and statistics
Software Development	Theory of Computation
Computer Organisation	Computer Networks
Advanced Algorithms	Convex Optimization for ML A^*

PROJECTS

Travelly

Jan'26 – Apr'26
Course Project: CS253, under Dr. Indranil Saha, CSE, IIT Kanpur

- Built a **trip management application** to streamline expense tracking, settlements and group logistics through **algorithmic debt simplification**, **UPI-based payment redirection**, **media management**, and **distance-minimizing route optimization**
- Utilized **Flutter** for the Frontend and **Node.js** for the backend of the application, with **PostgreSQL** for persistent data storage
- Implemented **Firestore-authenticated** API access with token verification caching, and backend user synchronization for invite-based group joining

Primality Testing & RSA Cryptosystem

Jan'25 – Apr'25
Course Project: CS648, under Dr. S. Baswana, CSE, IIT Kanpur

- Re-invented **Miller-Rabin primality test** using Fermat's Theorem and achieved $O(\log^3 n)$ time for primality validation
- Implemented **deterministic tests** for small inputs (Trial Division, Sieve of Eratosthenes) and **probabilistic tests** for large inputs (Fermat, Miller-Rabin) using **fast modular exponentiation**
- Built an **RSA Cryptosystem pipeline**: random prime generation via **Miller-Rabin**, key generation using **Euler's totient** and **Extended Euclidean algorithm**, and encryption/decryption using **modular exponentiation**

Building GemOS

Aug'25 – Nov'25
Course Project: CS330, under Dr. Debadatta Mishra, CSE, IIT Kanpur

- Built a **custom memory allocator** (malloc/free) using **mmap()** & implemented **mremap()** syscall for efficient memory remapping
- Implemented **kernel fault handling** for divide-by-zero faults, supporting instruction skip, function skip & user redirection
- Implemented **round-robin scheduling** for CPU allocation & performed multi-level **page table walks** for mapping inspection
- Achieved **multithreading** using clone/exit and semaphore based synchronization with **spinlocks** and **wait queues**

Hardware Implementation of Elliptic Curve Cryptography Apr'25

CS220 Case Study under Dr. Debapriya Roy, CSE, IIT Kanpur

- Designed the hardware architecture for a **Curve-25519 elliptic curve scalar multiplier** using Verilog HDL
- Designed a **finite field multiplier**, leveraging **modular reduction** techniques applicable to Solinas primes and an FSM architecture

PINN Application in Wave Simulation

May'24 – Jun'24
SURGE Project, under Dr. Subhajit Roy, CSE, IIT Kanpur

- Applied Physics Informed Neural Networks on 1D heat equation, modeling the **physics loss & data loss** using **mean squared error**
- Designed a model to predict the **thermal diffusivity constant**, while leveraging data loss to predict temperature distribution

MENTORSHIP EXPERIENCE

Contractual Instructor, Programming Pathshala

Jul'25
Conducted an **8-day Competitive Programming bootcamp** for students at Lovely Professional University

Academic Dept. Mentor, Electrical Engineering

Jul'24 – Aug'25
Mentored the juniors of my department in their **academic courses** and provided them with **career guidance**

Project Mentor, Deep Dive into Data Structures

May'24 – Jul'24
Mentored **35 mentees** to impart the knowledge of commonly used **data structures & algorithms** in Competitive Programming

A^* : GRADE FOR EXCEPTIONAL PERFORMANCE

Randomized Algorithms	Fundamentals of Computing A^*
Data Structures & Algorithms	Partial Differential Equations A^*
Intro to Stochastic Processes	Intro to Electrical Engineering A^*
Intro to Machine Learning	Principles of Programming Languages